15.S50 | January IAP 2015 | Graduate

# Poker Theory And Analytics

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## Instructor Insights

### Course Overview

This page focuses on the course *15.S50 Poker Theory and Analytics* as it was taught by Kevin Desmond during IAP 2015 .

This course takes a broad-based look at poker theory and applications of poker analytics to investment management and trading.

MIT Sloan News published an article on how Kevin Desmond links poker strategy to risk management in this course: *Game Theory: What Poker and Finance Have in Common*.

### Course Outcomes

#### Course Goals for Students

- Create an environment for study of poker theory without the need for real-money wagering.
- Develop the basic foundation for decision-making in poker.
- Allow students to assess their own level of play and have a framework for improvement.
- Provide an understanding of the current poker environment and how students might leverage talent for poker in the future.

#### Possibilities for Further Study/Careers

Management and leadership positions, finance, trade, global markets. The skills learned in this course are especially good for careers with high-pressure decision-making.

### Instructor Interview

> ***"I think there are a lot of ways for students to extract the benefits from poker without the need to risk money…"***

*—KEVIN DESMOND*

*In the following pages, Kevin Desmond describes various aspects of how he taught* 15.S50 Poker Theory and Analytics.

- Instructor Background and Motivation for Teaching the Course
- Gameplay, Connections and Contributors
- Challenges and Learning Opportunities

### Curriculum Information

#### Prerequisites

Permission of the instructor

#### Requirements Satisfied

Graduate subject credit

#### Offered

Offered during IAP when an instructor is available to teach the course

### Assessment

The students' grades were based on the following activities:

- 1/3 Gameplay: cash-in 1+ or play 10+ Daily Turbos
- 1/3 Casework: pass all three cases
- 1/3 Attendance: attend at least 8 of 11 lectures

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#### Instructor Insights on Assesment

Assessment in this course was an interesting puzzle. Poker is inherently a game where the results are not perfectly correlated with skill, but it is not in line with MIT's academic standards for course credit to be determined probabilistically. However, gameplay was an important learning tool and the students needed to be incentivized to play their best. The solution was to split into two leagues, one for credit, and one for prizes, where students would earn credit based on volume of play, but earn prizes based on success. This would ensure every student had the capacity to earn credit in reasonable time. In addition to gameplay, the homework assignments were key to assessing the students' understanding and ability to implement concepts taught during lectures.

### Student Information

#### Enrollment

About 150 students

#### Breakdown by Year

The breakdown of student in this course was about 60% graduate students, 40% undergrads.

#### Breakdown by Major

The graduate students were about 50% Management, 50% Engineering/Math/Science. The undergraduates were about evenly split among years and primarily from Engineering. A small number of students were visitors from other universities.

#### Typical Student Background

Students primarily had an interest in competitive games and practical applications of statistics. Some students intended to work in areas where poker knowledge was critical (trading or gaming industry), whereas other students had a more academic or recreational interest in poker. Most students had a strong background in statistics/math.

#### Ideal Class Size

The ideal class size is around 100-300 students. Since the class is heavily focused on gameplay, the goal is for students to be playing against a mix of familiar players and new opponents. By the end of the course, most students had a general level of familiarity of the different playing styles of students in the class.

### How Student Time Was Spent

During an average week, students were expected to spend between 19.5 and 24.5 hours on the course, roughly divided as follows:

#### In Class

The majority of in-class time was spent reviewing core concepts and solving example questions. Lecture attendance was mandatory for enrolled students and optional for listeners.

#### Out of Class

Outside of the classroom, a significant amount of time was spent playing poker in the online league, hosted by PokerStars. The average player played approximately 5000 hands of poker (the equivalent of about a year’s worth of live play). In addition, each of the three homework assignments required about 1-2 hours of work. For some students, about 2 hours per week was spent in review sessions.

### Course Team Roles

#### Instructor

Usually a graduate student with a background in professional poker.

#### Faculty Advisor

Faculty sponsor who generally oversees the course, but doesn’t necessarily have an active role in teaching sessions.

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## Instructor Background and Motivation for Teaching the Course

*In this section, Kevin Desmond describes his motivation for teaching 15.S50 Poker Theory and Analytics and how the course intersects with his professional and academic experiences.*

**Using Poker to Teach Key Business Concepts**

I spent several years playing poker professionally while studying finance as an undergraduate at Villanova University. I chose to join Morgan Stanley as a trader rather than pursue poker as a career. My transition into trading was very seamless as a result of my experience playing poker and interest in game theory concepts. I would like MIT students to have the same opportunity.

> ***"…concepts that are very difficult to teach new traders … are fundamental concepts covered throughout this type of poker education."***
>
> *—KEVIN DESMOND*

For students entering the business world, I believe the key concepts of poker that transfer well to management positions are decision making with incomplete information, reading the actions of others, and being comfortable with self-assessment. I believe these skills can benefit students in many aspects of life, not just their careers.

For those entering the trading field, in particular, experience with taking calculated risks, participating in a competitive global market, and familiarity with making decisions under pressure are concepts that are very difficult to teach new traders but are fundamental concepts covered throughout this type of poker education. I believe there are few activities that would prepare students for a career in finance better than poker.

**Creating an Opportunity for Students to Engage with Poker Conceptually, as Opposed to Gambling**

This class serves as an opportunity for students interested in trading and other careers to enjoy some of the benefits of learning and mastering poker without the need to risk money.

In fact, one of the primary motivators of the course was that it is difficult for students to engage with poker conceptually without the connection to gambling. This course allowed students to play poker competitively in a learning environment among those who have the same educational end goal. I think there are a lot of ways for students to extract the benefits from poker without the need for risking money and one of the core pillars of the course is encouraging students to avoid gambling when possible.

**From Wall Street to Sloan**

I enjoyed my time on Wall Street and I think it was a great opportunity to start my career in that type of environment. I found that at the end of the day, my greatest interest was in financial innovation and the opportunity to come to Sloan to learn among the most dedicated academics in the world was not one I could pass up.

I am excited for the opportunity to continue MIT's effort to be on the forefront of the effort to solve complex casino games. I was inspired by the MIT Blackjack team's ability to identify an opportunity and execute a sophisticated strategy so effectively, as depicted in the film *21*. I particularly enjoyed the emphasis they put on discovering a practical implementation of their analytical solution. I am a firm believer that an idea is only as valuable as its real-world viability and I am fortunate to be part of a university that encourages students to innovate with a practical end goal in mind.

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## Gameplay, Connections and Contributors

*In this section, Kevin Desmond describes the role of gameplay in 15.S50 Poker Theory and Analytics and offers advice for educators about actively linking concepts to real-world experience.*

> ***"…this course offers … the chance to see how the theory connects to the real world."***
>
> *—KEVIN DESMOND*

Gameplay was critical to each student's development in the course. In line with MIT's motto, *Mens et Manus*, I am a believer that practical application is the key to understanding fundamental concepts, and a robust, high volume gameplay league is the best way to provide that experience. Each lesson ends with practical "Rules of Thumb" to emphasize that all of the analytics have the end goal of aiding in the decision-making process at the table. I would recommend that other educators facilitating a similar course place a significant emphasis on structuring the league, as it serves as a key motivator and pedagogical tool for the course.

To other educators, I would also recommend that the course be actively linked to real-life poker as much as possible. There is an overwhelming amount of poker literature available to students, but a fundamental difference this course offers is the chance to see how the theory connects to the real world. The most successful lectures were ones in which we analyzed examples of live play or had guest speakers with decades-long poker careers talk about their experiences.

I was very surprised by what a positive impact outside contributions—in the form of software licenses, guest speakers, prize donations, etc.—had on the class. I was overwhelmingly impressed by the quality of guest speakers and how each one brought a completely unique life experience to the course. In addition, I owe a lot of the competitiveness of the online league to the supporters whose donations kept students interested in the league even after course credit was earned. If I were to teach the course again, I would put an even greater emphasis on outside contributions and integrate them more into the coursework. I believe there is a significant appetite for the type of audience that MIT students provide and I would like to increase the number of connections made between industry veterans and students.



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## Challenges and Learning Opportunities

*In this section, Kevin Desmond shares challenges he faced in teaching 15.S50 Poker Theory and Analytics and how he addressed them.*

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One challenge I faced in teaching this course was that the level of experience students had with poker varied substantially, ranging from students who could realistically play at a competitive level to those who still needed to learn the rules of poker. I decided to keep the lectures advanced, so that everyone would benefit from the time in class, and I quickly added several recitations during the week to help students who were struggling with concepts or needed a refresher on poker in general.

In planning for and teaching this course, I learned the value of being flexible. The few contingency plans I made were tested, thanks in part to a blizzard that struck Cambridge halfway through the course. There is no amount of planning that will prevent last-minute cancellations or hardware malfunctions and the best course of action, I learned, is to identify operational risks and adapt accordingly.

Teaching this course was also an opportunity for me to learn that there will never be enough time to teach everything there is to know about the subject. I found that spending more time on my areas of expertise was more interesting for the students and rewarding for me as an educator than trying to cover every topic in the limited amount of class time.